

AI Study Quiz Generator - Performance Report

Final Performance Score Metrics: 3/5 (60%)

Q1: What is the primary goal of Database Normalization according to the provided text?

Validated Correct Answer: To eliminate data redundancy and undesirable characteristics

Status Assessment: Incorrect

Context Trace: Database Normalization is the formal systemic approach of decomposing tables to eliminate data redundancy and undesirable characteristics like Insertion, Update, and Deletion Anomalies.

Q2: According to First Normal Form (1NF), what must be true about the values within all attribute domains?

Validated Correct Answer: They must contain only atomic, indivisible values

Status Assessment: Correct

Context Trace: First Normal Form (1NF) states that all attribute domains must contain only atomic, indivisible values.

Q3: What condition must be met for a relation to be in Second Normal Form (2NF), in addition to being in 1NF?

Validated Correct Answer: All non-prime attributes must be fully functionally dependent on the entire composite primary key

Status Assessment: Correct

Context Trace: Second Normal Form (2NF) mandates that the relation must already be in 1NF, and all non-prime attributes must be fully functionally dependent on the entire composite primary key, thereby eliminating partial dependencies entirely.

Q4: What specific type of dependency does Second Normal Form (2NF) aim to eliminate?

Validated Correct Answer: Partial dependencies

Status Assessment: Incorrect

Context Trace: Second Normal Form (2NF) mandates that the relation must already be in 1NF, and all non-prime attributes must be fully functionally dependent on the entire composite primary key, thereby eliminating partial dependencies entirely.

Q5: Third Normal Form (3NF) builds upon 2NF by ensuring that no non-prime attributes exhibit what kind of dependency on the primary identifier mechanism?

Validated Correct Answer: Transitive dependency

Status Assessment: Correct

Context Trace: Third Normal Form (3NF) advances this structure by verifying it holds 2NF compliance and ensuring that no non-prime attributes exhibit transitive dependencies on the primary identifier mechanism.